QQ-L-00171d(NAVY-Ships)
December 10, 1962
Interim revision of
Fed. Spec. QQ-L-171c
May 27, 1957

INTERIM FEDERAL SPECIFICATION

LEAD PIG

This interim Federal Specification was developed by the Bureau of Ships, Department of the Navy, Washington 25, D.C. based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above. The General Services Administration has authorized the use of this interim Federal Specification as a valid exception to Federal Specification QQ-L-171c.

1. SCOPE AND CLASSIFICATION

1.1 Scope.- This specification covers lead pigs of the composition specified in 3.3 (see 6.1).

1.2 Classification.-

1.2.1 Forms, shapes, and grades.- The pigs shall be furnished in commercial standard forms and shapes, in the following grades, as specified (see 3.3 and 6.2).

Grade A.
Grade B.
Grade C.

APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

NN-P-71—Pallets; Materials-Handling, Wood (General Construction Requirements).
QQ-S-781—Strapping, Flat; Steel.

Federal Standards:

FED-STD-102—Preservation, Packaging, and Packing Levels.
FED-STD-151—Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129—Marking for Shipment and Storage

FSC 9650

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(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, N. Y.)

3. REQUIREMENTS

3.1 Material.- The lead pigs shall be made from ore or other material by processes of reduction and refining. For grades A and B, reclaimed lead obtained by recovery of metallic lead and its alloys by the simple reclaiming process of melting, drossing, and casting is acceptable under this specification if chemical composition requirements are within the limits specified herein. Grade C lead pig shall be either primary or secondary lead produced from ore or other materials by a conventional process of reduction and refining or other processes which equally preclude the presence of unusual deleterious elements. Reclaimed lead is not acceptable for grade C.

3.2 Surface condition.- The pigs shall be clean and reasonably free from adhering foreign matter.

3.3 Chemical composition.-

3.3.1 Lead pigs shall conform to the chemical composition specified in table I.

Table I—Chemical composition

| 3/ Element | Grade A maximum | Grade B maximum | Grade C maximum |
|---|---|---|---|
| Lead | 1/ 99.90 | 1/ 95.0 | 1/ 99.90 |
| Antimony, tin and arsenic (total) | ---- | ---- | 0.002 |
| Iron | ---- | ---- | .002 |
| Bismuth | ---- | ---- | .025 |
| Zinc | ---- | ---- | .001 |
| Copper | ---- | ---- | 2/ .040 to .080 |
| Silver | ---- | ---- | .020 |

1/ Minimum by difference.
2/ Range.
3/ For Grade C, no element other than those specified shall exceed 0.001 percent maximum as determined by a normal spectrographic analysis.

3.3.2 The contractor shall furnish an analysis of each melt showing the percentage of each of the elements specified in 3.3.1.

3.4 Identification marking.- Each pig shall have the following information cast or stamped thereon in legible characters: The Federal Specification symbol, grade, the manufacturer's name or brand, and melt or cast number.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.



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4.2 Inspection lot.-

4.2.1 For inspection purposes a lot shall consist of all lead pigs of the same grade produced by one manufacturer and offered for delivery at one time, but shall not exceed one carload or fraction thereof.

4.2.2 For Grade C lead, a lot for chemical analysis shall also be limited to the lead pig produced by one manufacturing or smelting heat. In this case, a lot that cannot be identified by heat shall be cause for rejection.

4.3 Sampling for examination and tests.-

4.3.1 Sampling for visual examination.- A random sample of lead pig shall be selected from each lot offered for delivery and examined for visual characteristics in accordance with Standard MIL-STD-105 at inspection level I. The acceptable quality level shall be 2.5 percent defective (see 4.4.1).

4.3.2 Chemical analysis.- One pig shall be taken by the inspector to represent each ton of lead in the lot and the pig or pigs thus taken sampled by any one of the methods specified herein. Except when the spectrographic analysis is to be made on a solid sample (see 4.3.2.5), not less than 600 grams from each lot shall be forwarded to a laboratory for analysis (see 4.4.2).

4.3.2.1 Sawing.- The pigs shall be sawed completely through in the center, at each end of largest cross section, and at points midway between ends and center. The sawings shall be mixed thoroughly, quartered, and samples for analysis taken from the mixed materials. Sawings shall be free from all extraneous material and shall be treated with a strong magnet in order to remove iron introduced from the saw blade.

4.3.2.2 Drilling.- The pigs shall be drilled preferably with a 1/2-inch drill at least halfway through from two opposite sides. In drilling, the holes are spaced along a diagonal line from one corner of the pig to the other. If several pigs are used, the holes may be drilled in each of the several pigs of a group so that holes represent consecutive positions on the diagonal of a single pig. The drillings may be clipped and mixed and samples for analysis drawn from mixed material; or they may be melted in a clean crucible and either granulated by pouring into distilled water and drying thoroughly, or cast into thin slabs which shall be sawed through completely in several places and sawings treated as specified in 4.3.2.1.

4.3.2.3 Kettle (classes A and C).- Unless otherwise agreed upon by the procuring activity and the contractor, the pigs or portions of lot selected shall be melted in a clean kettle. The melt shall be made at as low a temperature as possible, preferably not over 685° F., in order to prevent excessive drossing, and the molten bath shall be stirred immediately prior to sampling. While stirring, a sample in a preheated ladle may be taken and cast into thin sample bars approximately 8 inches in length, 1-1/4 inches in width and 3/8 inch in thickness. From the bar so cast, saw cuts shall be made 1/2 inch apart alternately from each side, halfway across the slab. No cut shall be made closer than 1 inch to either end of the slab. The sample shall be treated with a strong magnet to remove iron.

4.3.2.4 When mutually agreed upon between the procuring activity and the contractor, ladle test ingots obtained during pouring of the metal shall be used as samples and treated as specified in 4.3.2.1, 4.3.2.2 and 4.3.2.3. One ladle test ingot shall be taken at the beginning of the pouring of a lot and one near the end of pouring as indicated by the inspector.

4.3.2.5 Spectrographic analysis.- Samples for spectrographic analysis shall be obtained by pouring metal representative of the heat or melt into a permanent mold in the presence of the inspector or at the time castings are poured. The mold shall be so designed that the samples obtained will be truly representative of the heat or melt and satisfactory for use by the laboratory at which the spectrographic analysis is to be performed.

4.4 Examination and tests.-

4.4.1 Visual examination.- Each of the sample lead pigs selected in accordance with 4.3.1 shall be visually examined to verify compliance with the requirements of this specification regarding surface condition and identification marking (see 3.2 and 3.4).

4.4.2 Chemical analysis.- Chemical analysis shall be made on each sample selected in accordance with 4.3.2 by wet chemical or spectrographic methods (method 111 or 112 of Standard FED-STD-151) to



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determine conformance with 3.3. In case of dispute, chemical analysis by wet chemical methods shall be basis for acceptance of chemical composition and spectrographic analysis the basis for the acceptance of other impurity elements as specified in table I, note 3/.

4.5 Rejected lots.- Any lead pig in the sample containing one or more defects shall be cause for rejection, and if the number of defective lead pigs in any sample exceeds the acceptance number for that sample, it shall be cause for rejection of the lot represented by the sample. Rejected lots may be offered again for inspection provided the contractor has repaired or removed all nonconforming lead pigs. Samples shall again be selected from such resubmitted lots and examined to verify compliance with this specification.

5. PREPARATION FOR DELIVERY

(For nonmilitary agencies, Standard No. 102 should be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment.)

5.1 Packing (see 6.2).-

5.1.1 Levels A and B.- Lead pig may be packed on pallets conforming to type III of Specification NN-P-71. The stringer ends shall be bevelled. The load shall be adequately tension tied to the pallet with 1-1/4-inch by 0.035-inch coated metal strapping conforming to Specification QQ-S-781. Full length, nominal 1 by 4 inch lumber shall be placed under the strapping at the top of the load. When pallets are not used, lead pig shall be packed in bundles stacked in a manner to facilitate handling with fork lift vehicles. Such bundles shall be adequately tension bound with 1-1/4 inch by .035 inch coated metal strapping conforming to Specification QQ-S-781. The gross weight shall not exceed 2,240 pounds.

5.1.2 Level C.- Lead pig, separated by grade, shall be packed in accordance with the supplier's commercial practice to assure receipt by carrier and safe delivery to the destination at the lowest transportation rate or rating. Containers shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation.

5.2 Marking.-

5.2.1 Military agencies.- In addition to any special marking required by the contract or order, loose pigs and palletized loads shall be marked in accordance with Standard MIL-STD-129.

5.2.2 Nonmilitary agencies.- Shipping containers shall be marked with the name of the material, the grade of pigs contained therein as defined by the contract or order under which shipment is made, the name of the contractor, the number of the contract or order, the specification symbol, and the Federal stock numbers.

6. NOTES

6.1 Intended use.-

6.1.1 Grade A.- Grade A lead pig is intended for foundry and chemical uses.

6.1.2 Grade B.- Grade B lead pig is intended for uses such as weights and ballast.

6.1.3 Grade C.- Grade C lead pig is intended for structural uses where high purity lead is required.

6.2 Ordering data.- Purchasers should exercise any desired options offered herein, and should specify the following (see 4.3.2 and 4.3.2.4):

(a) Title, number, and date of this specification.
(b) Grade of lead pig required (see 1.2).
(c) Whether level A, B or level C packing is required (see 5.1).

Notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise, as in any manner licensing the holder or any other person or



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**SPECIFICATION ANALYSIS SHEET**
**NAVSHIPS-4863 (8-61)**

INSTRUCTIONS

**BUDGET BU. NO. 45-R309**

*This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Bureau of Ships*

*This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost.*

*Comments and the return of this form will be appreciated.*

*Fold on dotted lines on reverse side, staple in corner, and send to Bureau of Ships, Specifications and Standardization Branch, Washington 25, D.C.*

SPECIFICATION

| ORGANIZATION | CITY | STATE |
|---|---|---|
| | | |

| CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT |
|---|---|---|
| | | $ |

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   a. GIVE PARAGRAPH NUMBER AND WORDING

   b. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID.

3. IS THE SPECIFICATION RESTRICTIVE? ☐ YES ☐ NO IF THE ANSWER IS "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification.)* PLACE THIS FORM AND PAPERS IN AN ENVELOPE AND SEND TO THE BUREAU.

| SUBMITTED BY *(Print name and activity)* | DATE |
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